KPI REQUIREMENTS

Interns must calculate the following KPIs:

Population KPIs

1. **Total Penguins Count**
    - Total number of valid records after cleaning.
2. **Species Distribution**
    - Count of penguins by species
    - Percentage contribution by species
3. **Island-wise Population**
    - Total penguins per island
    - % distribution by island

Biological Measurement KPIs

4. **Average Body Mass (g)**
    - Overall average
    - Average by species
    - Average by gender
5. **Average Flipper Length (mm)**
    - Overall average
    - By species
    - By gender
6. **Average Bill Length (mm)**
    - Overall average
    - By species
7. **Average Bill Depth (mm)**
    - Overall average
    - By species

8. **Bill Ratio KPI**
    - Bill Length / Bill Depth
    - Average ratio by species

Gender-Based KPIs

9. **Male vs Female Count**
    - Total males
    - Total females
    - Percentage split
10. **Size Difference by Gender**

- Avg body mass difference (Male vs Female)
- Avg flipper length difference

Correlation KPIs

11. **Correlation Matrix**

- Body mass vs flipper length
- Bill length vs bill depth

12. **Highest Positive Correlation Pair**

- Identify strongest relationship among numeric features

Island-Level KPIs

13. **Species per Island**

- Species count breakdown by island

14. **Average Body Mass per Island**
15. **Average Flipper Length per Island**

Data Quality KPIs

16. **Missing Value Percentage**

- Per column

17. **Outlier Detection (Optional Advanced)**

- Body mass outliers
- Flipper length outliers

CHART REQUIREMENTS

Interns must build the following visualizations:

Distribution Charts

1. **Species Distribution (Bar Chart)**
    - X-axis: Species
    - Y-axis: Count
2. **Island Distribution (Bar Chart)**
3. **Gender Distribution (Pie or Donut Chart)**

Measurement Distribution Charts

4. **Body Mass Distribution (Histogram)**
5. **Flipper Length Distribution (Histogram)**
6. **Bill Length Distribution (Histogram)**

Comparison Charts

7. **Average Body Mass by Species (Bar Chart)**
8. **Average Flipper Length by Species (Bar Chart)**
9. **Average Body Mass by Gender (Bar Chart)**
10. **Species Distribution by Island (Stacked Bar Chart)**

Relationship Charts

11. **Flipper Length vs Body Mass (Scatter Plot)**

- Color by species
- Size by bill length (optional)

12. **Bill Length vs Bill Depth (Scatter Plot)**
13. **Correlation Heatmap**

Executive Dashboard Requirements

Dashboard must include:

Filters (Real-Time)

- Species (Multi-select)
- Island (Multi-select)
- Gender (Multi-select)

KPI Cards (Top Section)

- Total Penguins
- Avg Body Mass
- Avg Flipper Length
- Avg Bill Length
- Male vs Female %

Charts Section

- 1 Distribution chart
- 1 Comparison chart
- 1 Relationship chart
- 1 Correlation heatmap